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Decoding Sustainability Communication via NLP

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Introduction

- Sustainability websites = transparency + persuasion
- Signal legitimacy and engage stakeholders
- Shell = technical; WWF = emotional
- Manual coding lacks scale/nuance
- **Aim:** Use NLP to compare Shell vs WWF

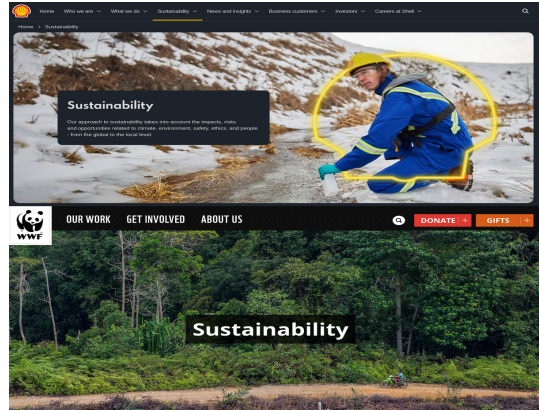


Diagram: corp vs NGO

Stakeholder Theory Expectations

- Message shaped by stakeholders
- Shell: investors/regulators = "investment", etc.
- WWF: donors/public = "donor", "nature"
- Shell: data-driven; WWF: community focus
- Predictable via lexical/semantic analysis

Type	Term
Economic, Regulatory	investment
Economic, Regulatory	emission
Economic, Regulatory	energy
Community, Environment	donor
Community, Environment	conservation

Graphic: stakeholder mapping

Framing Theory Expectations

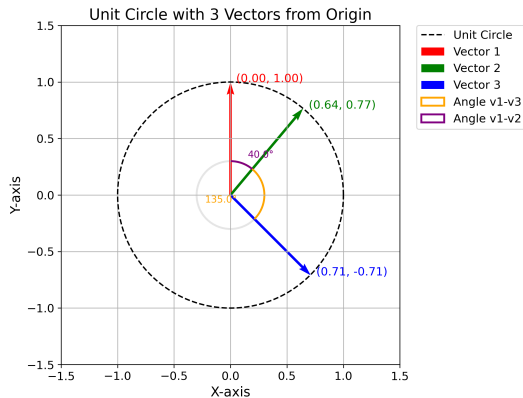
- Framing guides perception
- Shell: optimistic "innovation" frame
- WWF: moral/problem frame ("action", "duty")
- Shell: business advantage; WWF: urgency
- Detectable via framing terms

Type	Term
Opportunity	innovation
Opportunity	technology
Moral	duty
Moral	crisis
Moral	action

Diagram: framing strategies

Data & Methodology

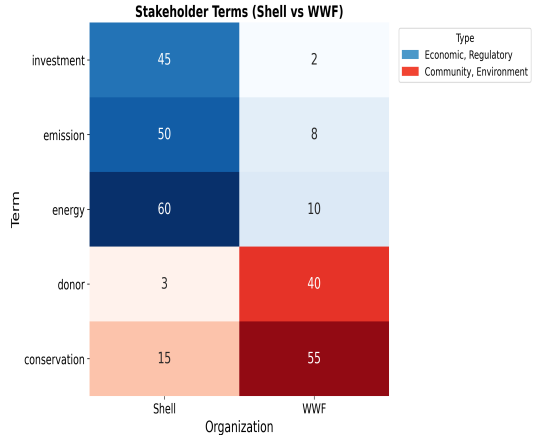
- 7 webpages each from Shell and WWF
- Cleaned HTML, kept main content
- SpaCy tokenization, lemmatization
- Lexical: count theory-based keywords
- Semantic: BERT embeddings + similarity



NLP pipeline sketch

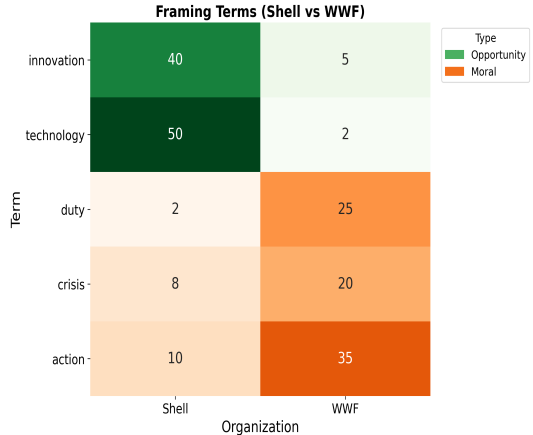
Lexical Analysis Results (1)

- Shell: "energy", "emission", "investment"
- WWF: "conservation", "donor", "nature"
- Terms rarely overlap between orgs
- Shell = tech & opportunity; WWF = duty & action
- Clear lexical patterns match theory



Lexical Analysis Results (2)

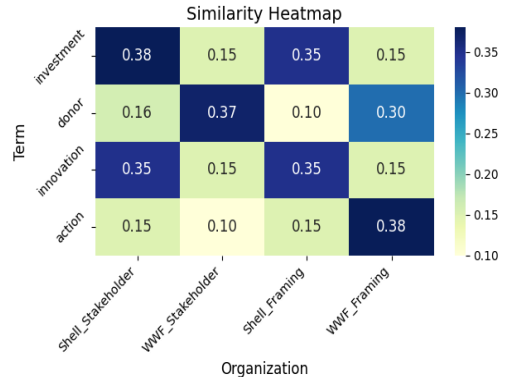
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Frequency bar chart

Semantic Similarity (BERT)

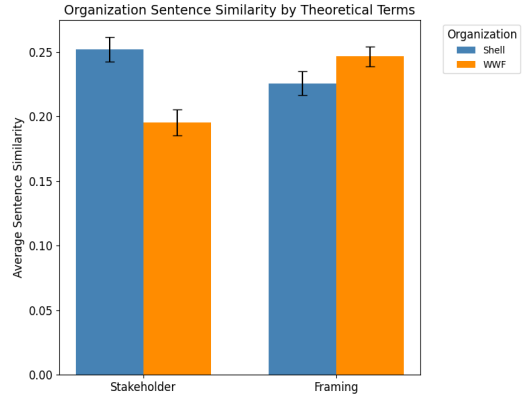
- Shell aligns with "investment", etc.
- WWF aligns with "donor", "action"
- Shell avg. stakeholder similarity = 0.26
- WWF avg. framing similarity = 0.25
- Semantic match confirms lexical trends



Similarity heatmap

Theoretical Implications

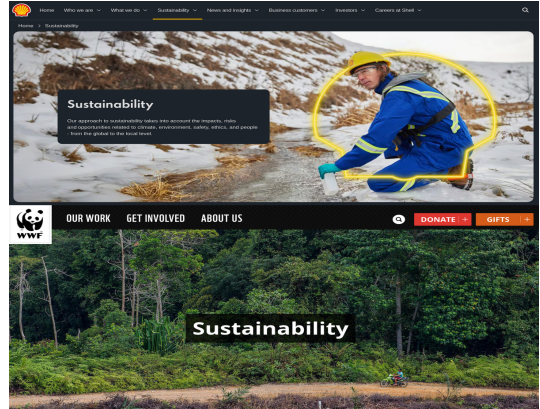
- Shell confirms stakeholder focus
- WWF shows moral advocacy focus
- Shell = opportunity frame; WWF = moral frame
- NLP helps refine both theories
- Empirical grounding boosts explanatory power



Concept map: theory + findings

Communication Implications

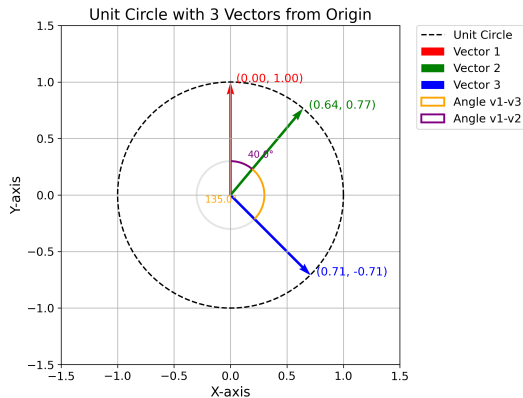
- Shell = competence; WWF = emotional appeal
- Tailored messaging for each audience
- NLP = replicable insight, not anecdotal
- Firms = clarify tech; NGOs = build urgency



Icons: tech vs moral appeals

Scalable Analysis

- Method scales to 100s of sites
- Cloud + BERT = efficient workflows
- Topic models could reveal trends
- Text + image data can be fused
- Big data boosts theory + practice



Cloud or NLP infrastructure diagram

Conclusion & Future Directions

Multimodal analysis diagram

- NLP reveals Shell vs WWF messaging logics
- Confirms stakeholder + framing theory
- Lexical + semantic = stronger analysis
- Actionable insights for communication design
- Future: multimodal + large-scale studies